Weeks 5–6: Project Initialization and First Developments

Objective of these two weeks:

Move from design to implementation: initialize the Django project, implement admin-controlled JWT authentication, and validate the technical stack before starting the Core Platform development.

Week 5 (from 02/03/2026 to 06/03/2026) Environment Setup and Project Initialization

During the fifth week of my internship, I moved from the design phase to the implementation phase by setting up the development environment for the project. I installed and configured the main technologies required for the platform, including Django for the backend, React for the frontend, and PostgreSQL as the database system. I initialized the Django project and configured environment variables to ensure that sensitive information such as database credentials and secret keys are not stored directly in the source code. I also structured the project to support a clean development workflow between the frontend and backend components. In addition, I created and configured the GitHub repository for the project in order to manage version control and keep the development process organized. During this phase, I also discussed some technical choices with my supervisor and a member of the Test Factory team at BIAT IT, who provided useful advice regarding the development approach and the structure of the platform. This phase allowed me to better understand how to prepare a professional development environment and ensure that all components of the stack communicate correctly. Setting up the environment correctly at this stage was important because it establishes the technical foundation on which the rest of the platform will be built.

Week 6 (from 09/03/2026 to 13/03/2026) User Model and Authentication System

During the sixth week of my internship, I focused on implementing the first functional component of the platform, which is the user management and authentication system. I started by developing a custom User model adapted to the BIAT organizational structure. The goal was to ensure that user accounts are controlled internally and cannot be created freely by anyone on the platform. In this system, only administrators at BIAT are allowed to create user accounts. When a new user is created, the administrator generates the account and provides the user with their credentials, including a password and a professional email address following the format *name.surname@biat-it.tn*. Users can then log in either using their generated email address or their matricule together with their password. To implement secure authentication, I set up a JWT-based authentication system using Simple JWT in Django REST Framework. I also began developing a dedicated admin dashboard for BIAT administrators and team leads, which allows them to manage user accounts inside the platform. This dashboard is separate from Django's built-in admin panel, which remains reserved for me as the platform owner for system-level administration and maintenance. One difficulty I encountered during this phase was building the Axios service on the React side in order to properly communicate with the Django backend and handle JWT authentication. I had to configure the Axios instance to correctly send authentication tokens in the request headers and manage secure communication between the frontend and backend. Through this work, I gained a deeper understanding of how to adapt Django's authentication system to real organizational requirements and how to design an access control structure that separates business-level administration from technical platform management.